

Suppose we have three points in the original image:

$$(0, 0), \quad (1, 0), \quad (0, 1)$$

and we know where they ended up in the second image:

$$(2, 3), \quad (4, 3), \quad (2, 6)$$

We want to find the affine transformation

$$x' = ax + by + t_x$$

$$y' = cx + dy + t_y$$

There are **6 unknown parameters**:

$$a, b, c, d, t_x, t_y.$$

1. Use the first point

For $(x, y) = (0, 0) \rightarrow (2, 3)$:

$$a(0) + b(0) + t_x = 2$$

$$c(0) + d(0) + t_y = 3$$

so

$$t_x = 2, \quad t_y = 3.$$

1. Use the first point

For $(x, y) = (0, 0) \rightarrow (2, 3)$:

$$a(0) + b(0) + t_x = 2$$

$$c(0) + d(0) + t_y = 3$$

so

$$t_x = 2, \quad t_y = 3.$$

2. Use the second point

For $(1, 0) \rightarrow (4, 3)$:

$$a(1) + b(0) + 2 = 4$$

so

$$a = 2.$$

And

$$c(1) + d(0) + 3 = 3$$

so

$$c = 0.$$

3. Use the third point

For $(0, 1) \rightarrow (2, 6)$:

$$a(0) + b(1) + 2 = 2$$

so

$$b = 0.$$

And

$$c(0) + d(1) + 3 = 6$$

so

$$d = 3.$$

Therefore our affine transformation is

$$\begin{aligned}x' &= 2x + 2 \\y' &= 3y + 3\end{aligned}$$

It scales x by 2, scales y by 3, and translates by $(2, 3)$.

Why do we need 3 matches?

This is the key teaching point.

Each point correspondence gives us **two equations**:

$$(x, y) \rightarrow (x', y')$$

gives one equation for x' and one for y' .

An affine transformation has **6 unknowns**:

$$a, b, c, d, t_x, t_y.$$

So we need

$$\frac{6 \text{ unknowns}}{2 \text{ equations per match}} = 3 \text{ point matches.}$$

Affine transformation: 6 unknown parameters

Each point match: 2 equations

Minimum: 3 point matches